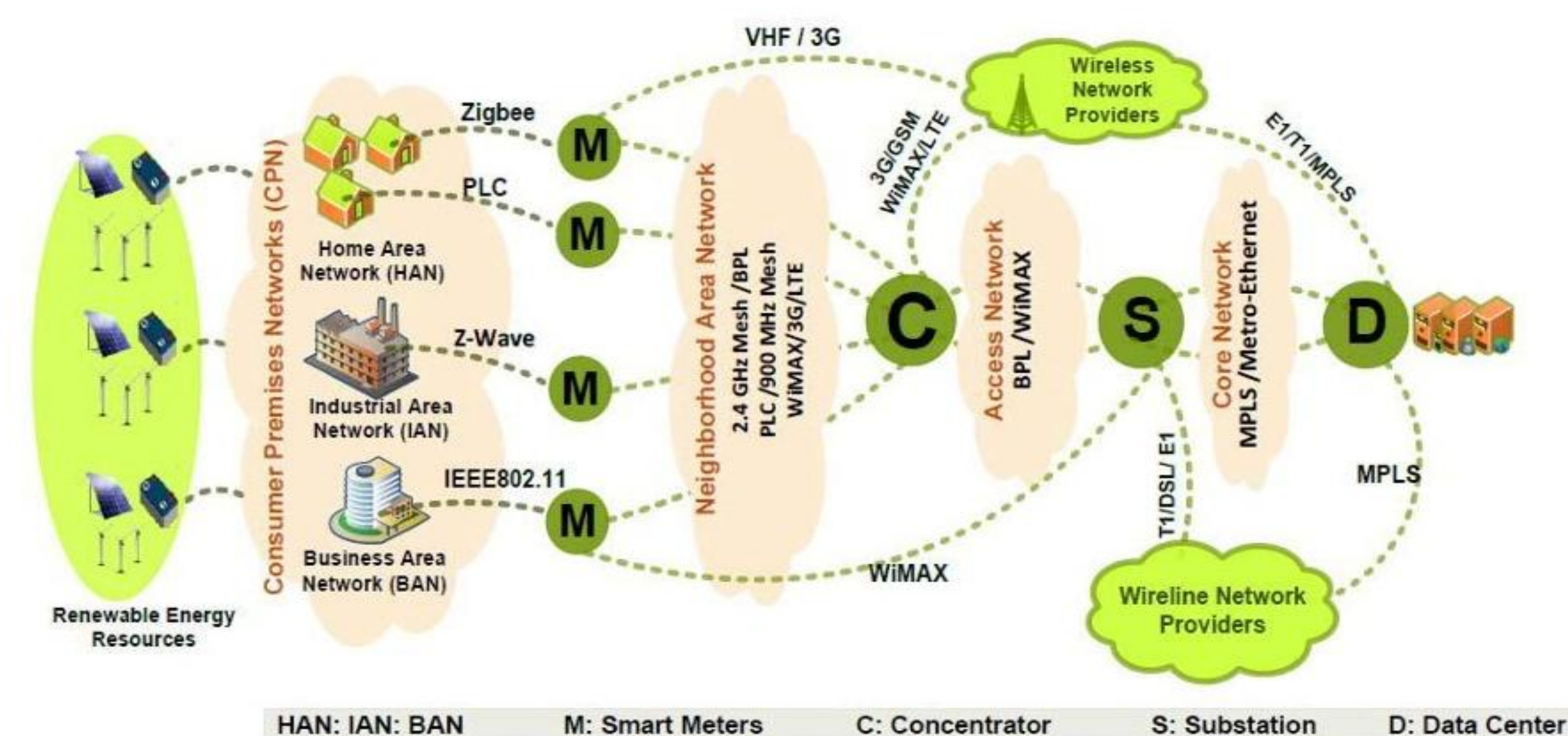


INTRODUCTION

- Smart grids integrate digital communication with power systems
- Smart meters enable real-time monitoring and control
- Communication networks introduce cybersecurity risks
- Securing these systems is essential for reliability

SYSTEM ARCHITECTURE

- Smart grids use layered communication systems for efficient data flow
- Smart meters collect and transmit electricity usage data
- Data flows through network layers to centralized control systems
- Utility servers process data for monitoring and billing



SECURITY ISSUES IN SMART METER COMMUNICATION

Data Integrity

- Unauthorized modification of meter data
- Leads to incorrect billing and unreliable system decisions

Confidentiality

- Sensitive user data can be intercepted
- Results in privacy breaches

Availability

- Disruption of communication channels
- Can affect grid operations and service continuity

Authentication

- Weak access control mechanisms
- Allows unauthorized system access

COUNTER MEASURES

Encryption

- Secures data during transmission
- Prevents unauthorized access

Authentication Mechanisms

- Verifies identity of devices and users
- Prevents unauthorized entry

Monitoring & Detection

- Detects unusual network activity
- Enables early threat identification

Secure Communication Protocols

- Ensures safe and reliable data exchange
- Reduces system vulnerabilities

IMPACT OF SECURITY BREACHES

- Incorrect billing and financial loss
- Leakage of sensitive consumer data
- Disruption of grid operations
- Loss of trust in smart grid systems

CONCLUSION

- Smart grids improve efficiency but introduce cybersecurity risks
- Communication systems are key vulnerability points
- Strong security measures are essential for reliable operation

Case Study: Attempted Cyber Reconnaissance on Swedish Energy Infrastructure

Context

- Swedish energy infrastructure
- Attempted cyber intrusion (early-stage attack)

Motivation

- Disrupt critical infrastructure: Probe Sweden's power grid to find weaknesses that could disrupt electricity supply
- Create economic / operational instability: Test access paths that could lead to operational and financial instability
- Geopolitical influence: Demonstrate cyber capability to exert geopolitical pressure on a European nation

Attack Vector

- Network scanning of exposed systems
- Identifying vulnerabilities
- Attempt to access control systems

Potential Impact

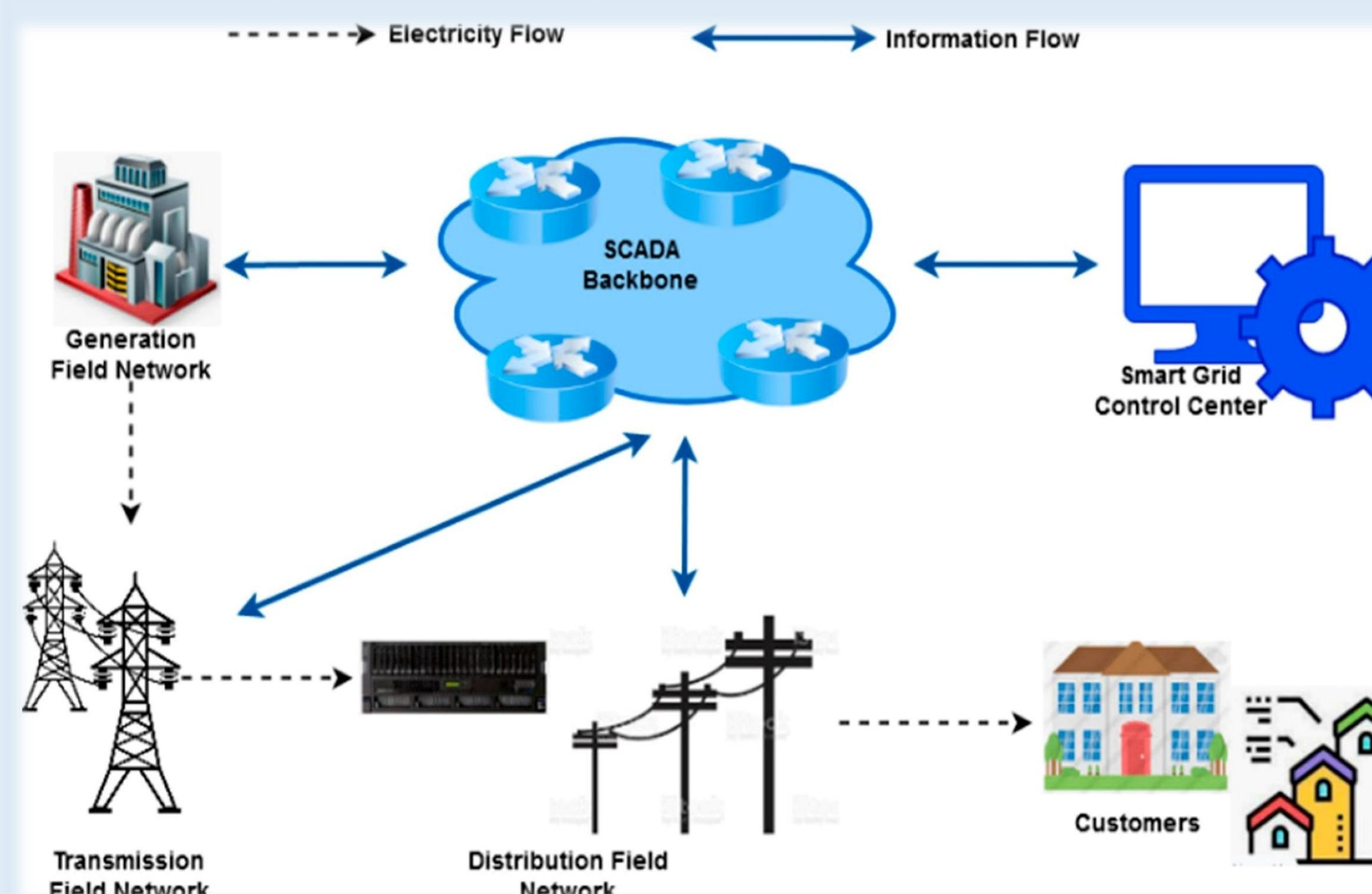
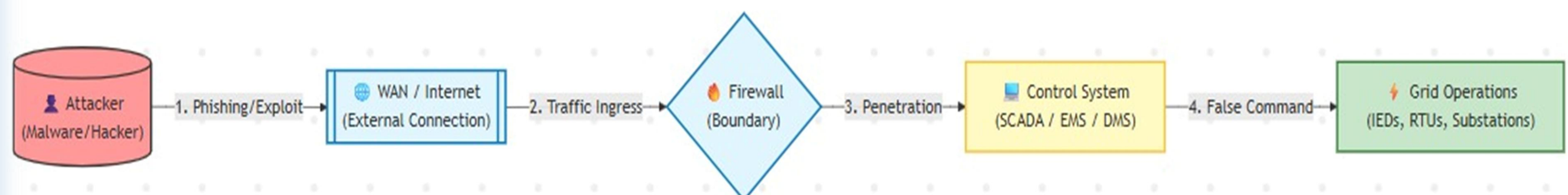
- Power disruption
- Incorrect data / control signals
- Financial and operational damage

Why It Failed

- Early detection, Monitoring systems, Network segmentation

Key Lessons

- Early-stage attacks are critical
- Monitoring is essential
- Communication layers are vulnerable
- Strong authentication is required



Public-Facing Network (Internet/WAN)

External Access Points (Remote Interfaces)

Perimeter Systems (Firewall-facing assets/gateways)

Monitoring/Detection Layer

- Generation Field Network** - Not directly targeted in this case
- Transmission Field Network** - Indirectly dependent on SCADA (potential downstream risk)
- Distribution Field Network** - Could be impacted if SCADA is compromised
- SCADA Backbone (CORE SYSTEM)** – Primary Target in the attack attempt
- Smart Grid Control Center** - Only affected if upstream systems fail
- Customers (End Users)** - High-value target, but NOT reached

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